INSIDE STUDIO SP · VST3 · AU

MESA

ANALOG CONSOLE

Step-by-step manual

Version 0.5.1 · Windows and macOS · VST3 and AU



insidestudiosp.com.br/mesa · contato@insidestudiosp.com.br

What is MESA

MESA is an **analog console** inside your DAW. It comes as **two plugins** in the same installer:

- **MESA CANAL**: goes at the **end of every track**. It processes that track with a preamp (PRE), compressor (DYNAMICS), 4-band EQ, pan and fader.
- **MESA** (the console): goes on the **master**. Its window is the whole mixer: all channels side by side, the groups, the main meter, the master COLA BUS and the group and master faders.

The **audio follows your DAW's signal path**: each MESA CANAL processes its own track, so sends, returns and groups keep working. Turning a knob on the console turns it on that track's MESA CANAL, and your DAW records the automation.

The circuits carry the tone of classic consoles: **E-4K**, **API 312** and **BRIT 73** preamps; **E-4K**, **BRIT 81** and **API 550** EQs; **VCA 4K**, **COLA 4K**, **FET 76** and **OPTO 2A** compressors.

Before you start

ITEM	DETAILS
Format	VST3 (Windows and macOS) and AU (macOS). Windows 10 or 11, 64-bit; macOS 11 Big Sur or newer.
DAWs	The console sees the channels when the DAW loads the plugins **in the same process**, which is the default in Ableton, Logic, Reaper, FL Studio and Studio One.
DAW fader	Leave the DAW fader at **0 dB** and use the MESA fader.

Installation and activation

1. Download the installer from the link in your purchase e-mail and choose Windows or Mac.
2. Run the installer. It installs both plugins: **MESA** and **MESA CANAL**.
3. Open your DAW and, if needed, have it scan for new plugins (rescan).
4. The first time, enter the **e-mail** and the **license token** you received.

> **LICENSE.** The same token works on up to 2 computers. To switch machines, deactivate it on the old computer and the slot is freed. Internet is only needed to activate: after that it works up to 45 days offline and renews by itself when you're back online. To see the license: plugin menu › `Licença…` (license).

> **NOT ACTIVATED**, the plugin opens but passes the sound straight through, unprocessed.

STEP 1

1 MESA CANAL at the end of every track

1. Put **MESA CANAL** as the **last effect** on every track (after your other plugins).
2. At the top of the window, leave it on `CANAL` (channel; it's the default).
3. On the DAW's **group channels** (drums, rhythm, vocals…), also add a MESA CANAL and choose `GRUPO` (group).

Each MESA CANAL processes only its own track. Its window shows the full channel strip (on the left) and, next to it:

- **CONSOLE**: the green light and "Ligado à MESA no master" (linked to MESA on the master) confirm the console found this channel;
- **POSIÇÃO NA CONSOLE** (console position): the number (in red), the name and the group (G1, G2 or G3);
- **MEDIDORES** (meters): INPUT, GAIN REDUCTION and OUTPUT VUs;
- **CARÁTER** (character): the PRE, EQ and DYN models and the ORDEM (order: EQ → DYN or DYN → EQ).

> **ABLETON.** Ableton doesn't pass the track name to the plugin. MESA CANAL reads the name from its own window title: **open each track's MESA CANAL window once** and the name is saved in the session.



MESA CANAL window

STEP 2

2 MESA (console) on the master

Put **MESA** on the DAW master and open its window. At the bottom of the sidebar, the green light shows `no master` (on the master), and **CANAIS** (channels) shows how many are connected (e.g. "15 conectados · 12 canais · 3 grupos": 15 connected · 12 channels · 3 groups).

STEP 3

3 Order, name and group of each channel

Order on the console

Channels appear in **the DAW's track order**: MESA uses the number the DAW reports, or the number at the start of the track name ("3-Audio"). The number shows as a red LED on top of each channel.

- To set the position by hand: **double-click the red number**, or MESA CANAL menu › `Posição na console…` (console position)
- To rename: **double-click the channel name**, or menu › `Renomear este canal…` (rename this channel)

Groups G1, G2 and G3

MESA has 3 groups. A MESA CANAL set to `GRUPO` becomes group G1, G2 or G3 (pick it on the AUX row or in "vai para": goes to). The group track's name becomes the group name in the center section (e.g. BATERIA, BASE, VOZES).

On regular channels, G1/G2/G3 marks the channel's group (the top display shows it, e.g. `G1 · BATERIA`).

> **IMPORTANT.** Audio reaches the group through **your DAW's routing**. MESA doesn't change the signal path: route the track to the group channel in your DAW, as usual.

STEP 4

4 Get to know the console



- **Left**: the sidebar (console options, zoom, meter, quality, LIMPA SOLO).
- **Middle**: the mixer, channels side by side. Each channel has, top to bottom: CANAL (models and number), INPUT, DYN, EQ, AUX and OUT (pan, SOLO, CUT and fader).
- **Top right**: the selected channel's display (number, name, group) with ◀ ▶ arrows to step through channels.
- **Right**: the SEÇÃO CENTRAL (center section: MASTER meter, gain reduction VUs, COLA BUS and the group and master faders).

STEP 5

5 INPUT: the preamp (PRE)

Click the PRE name (the button next to PAD) to switch models:

MODEL	CHARACTER
E-4K	British console preamp: clean and solid, a touch of 2nd harmonic, 3rd only at the top.
API 312	Discrete op-amp: hard knee, 3rd with a bit of 2nd, punchy transients.
BRIT 73	Class A: 2nd harmonic early on; the low end saturates first in the transformer and gets fatter.

CONTROL	RANGE · DEFAULT	WHAT IT DOES
GAIN	−20 to +20 dB · **0 dB**	Level going into the channel.
DRIVE	0 to 10 · **0**	Pushes the signal into the circuit (up to +20 dB). Volume stays similar: it changes the character (saturation).
PAD	on/off · **off**	Cuts 20 dB at the input.
HPF	20 to 400 Hz · **OFF**	18 dB/octave high-pass.
LPF	3 to 20 kHz · **OFF**	12 dB/octave low-pass.

STEP 6

6 DYN: the compressor

Click the DYNAMICS **green display** to switch models. The LEDs show gain reduction.

MODEL	CHARACTER
VCA 4K	The British console's channel compressor, clean. ATTACK 0.1 to 30 ms. RELEASE fully up = AUTO.
COLA 4K	The same console's bus compressor, stepped: ATTACK 0.1 / 0.3 / 1 / 3 / 10 / 30 ms and RELEASE 0.1 / 0.3 / 0.6 / 1.2 s or AUTO.
FET 76	ATTACK 20 to 800 µs, RELEASE 50 ms to 1.1 s, RATIO 4/8/12/20 and ALL. Gets dirtier the harder it compresses.
OPTO 2A	Optical tube leveler: wide knee, two-stage RELEASE. Low RATIO = COMP, high = LIMIT.

CONTROL	RANGE · DEFAULT	WHAT IT DOES
THRESH	−40 to +10 dB · **OFF**	Point where the compressor starts working. Fully up, it's off.
RATIO	1:1 to 20:1 · **4:1**	How much it compresses.
ATTACK / RELEASE	0 to 10 · **6** / **AUTO**	The actual time depends on the model (hover to see it in ms).
GATE	−80 to −10 dB · **OFF**	Closes the channel below the chosen level.
DRY/WET	0–100% · **100%**	Blends the compressed sound with the original (parallel compression).
PRE	on/off · **off**	On: dynamics come before the EQ (DYN → EQ).

STEP 7

7 4-band EQ

Click the EQ name (next to "EQ") to switch models. `IN` turns the EQ on and off.

MODEL	CHARACTER
E-4K	Q knob on the mids; dark faceplate.
BRIT 81	Stepped frequencies, normal/HIGH Q, inductor; gray faceplate.
API 550	Stepped frequencies and gain; Q narrows as gain increases.

BAND	FREQUENCY · DEFAULT	CONTROLS
HF	1.5 to 16 kHz · **8 kHz**	GAIN (±15 dB) and FREQ. `BELL` switches shelf to bell.
HMF	600 Hz to 7 kHz · **2.5 kHz**	GAIN, FREQ and Q (0.4 to 4).
LMF	200 Hz to 2.5 kHz · **500 Hz**	GAIN, FREQ and Q (0.4 to 4).
LF	30 to 450 Hz · **100 Hz**	GAIN and FREQ. `BELL` switches shelf to bell.

> **TIP.** Hover over a knob to see the value the circuit actually uses (the BRIT 81 and API 550 step, the compressor time in ms).

STEP 8

8 AUX and OUT: group, pan and fader

- **G1 / G2 / G3** (AUX row): the channel's group.
- **PAN**: from L100 to R100; center is C.
- **SOLO**: a soloed channel mutes the other **channels** (not the groups). `LIMPA` (clear), in the sidebar, clears all solos.
- **CUT**: mutes the channel.
- **Fader**: from +10 dB to −∞, with an LED meter beside it. Below, the fader level and the meter level.



AUX and OUT of a channel

Signal path order

At the head of each channel, the three green displays show the processing order and the model of each stage (e.g. PRE 73 · EQ 4K · DYN 76).

STEP 9

9 Center section and COLA BUS



- **MASTER**: main meter with PEAK and RMS (in dBFS, with the max below) and goniometer (L/R). `ZERAR` (reset) clears the max values.
- **Bus tabs** (the 3 groups and MASTER) and the **gain reduction VUs**: click a tab or a VU to pick the bus. The big VU shows its gain reduction.
- **Faders** for the 3 groups and the master, at the bottom.

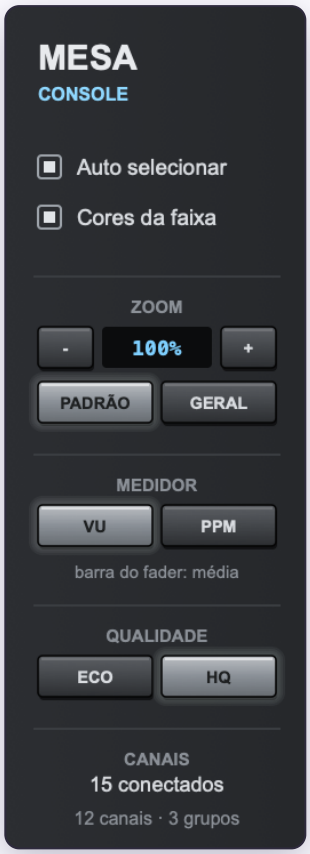
COLA BUS (master compressor)

CONTROL	RANGE · DEFAULT	WHAT IT DOES
THRESHOLD	−30 to +6 dB · **0 dB**	Point where compression starts.
MAKEUP	0 to +20 dB · **0 dB**	Restores level after compression.
DRY/WET	0–100% · **100%**	Parallel compression.
ATTACK	0.1 / 0.3 / 1 / 3 / 10 / 30 ms · **10 ms**	Stepped, like the original.
RELEASE	0.1 / 0.3 / 0.6 / 1.2 s / AUTO · **AUTO**	Stepped.
RATIO	1.5 / 2 / 4 / 10 : 1 · **4:1**	Stepped.
IN	· **on**	Turns the COLA BUS on and off.
SC HPF	· **off**	Filter on the detector: the low end stops "pulling" the compression.

With a group tab selected, the section shows that group's compressor (the DYNAMICS of the group's MESA CANAL).

Sidebar, zoom and mouse

OPTION	WHAT IT DOES
Cores da faixa	(Track colors) Paints each channel with its track color.
ZOOM − / +	Mixer zoom from 50% to 200%. Click the number to go back to 100%.
PADRÃO / GERAL	(Default / Overview) GERAL shrinks the mixer to show all channels at once.
MEDIDOR VU / PPM	(Meter) The bar next to the fader shows average (VU) or peaks (PPM).
QUALIDADE ECO / HQ	(Quality) HQ: saturation with 4× oversampling (default). ECO uses less CPU. Applies to all channels.
LIMPA SOLO	(Clear solo) Removes solo from all channels.



Collapsible rows

Click the row label on the left (**CANAL, INPUT, DYN, EQ, AUX**) to collapse or expand that row on every channel. The freed space goes to the fader. It's saved in the session.



Zoom at 140% with the INPUT and DYN rows collapsed.

Mouse

- **Drag** the knob or fader; hold **Shift** for fine adjustment.
- **Double-click** a knob or fader: back to default. On the name: rename. On the red number: change position.
- On the console, the **mouse wheel** scrolls through channels; **Ctrl + wheel** changes the zoom.

FAQ and troubleshooting

SITUATION	WHAT TO DO
A channel doesn't show up on the console	Check that the track has MESA CANAL and that its window shows "Ligado à MESA no master". The DAW must load plugins in the same process (no plugin isolation/sandboxing).
In Ableton, the channel has no name	Open that track's MESA CANAL window once: the name is read and saved in the session.
Channels are out of order	Double-click the red number (or menu › Posição na console…) and choose the position. Or start the track name with the number ("3-Audio").
The sound didn't change at all	Check that the license is active (menu › Licença…). Not activated, the sound passes straight through.
The group doesn't receive the channel	Sending to the group is done in the DAW's routing. G1/G2/G3 in MESA only tags the group on the console.
The volume differs from the DAW	Leave the DAW fader at 0 dB and use the MESA fader.
High CPU	In the sidebar, switch QUALIDADE to ECO.
I want to reset a knob to default	Double-click it.

30-second summary

1. **MESA CANAL** at the end of every track (CANAL) and on group channels (GRUPO + G1/G2/G3).
2. **MESA** on the master: open the window and see the whole console.
3. DAW fader at 0 dB; use the MESA one.
4. Choose the character: PRE, EQ and DYN on each channel.
5. Adjust INPUT, DYN, EQ, pan and fader on the console.
6. Finish on the master COLA BUS.

> **SUPPORT.** Questions, suggestions or issues: **contato@insidestudiosp.com.br** · **insidestudiosp.com.br**